Mondrian and Residual-Weighted Conformal Uncertainty Bands for Physics-Informed American Option Pricing

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***Abstract*—Physics-informed neural networks (PINNs) can price options without labelled prices, but a point price is not a risk number. We study split conformal prediction on top of PINN pricers for European and American puts under geometric Brownian motion, with exact Black-Scholes and 500-step binomial prices as calibration and test labels. Over ten training seeds, the PDE loss cuts the European mean absolute error from \$4.42 (a control without the PDE term) to \$0.25 and shrinks the 90% split-conformal band from \$19.43 to \$0.94. When 3000 labels are available, eight of ten labelled regressors beat the PINN inside their training region, whereas kernel and polynomial ridge fail once they leave that region. Split conformal still hits nominal marginal coverage for every model. Strengthening a weak complementarity penalty reduces the American PINN's bias to \$0.22 and at the same time raises calendar violations (16.8% of pairs). The raw PDE residual carries almost no signal for the American put (rank correlation 0.01) and widens the band, whereas the complementarity residual $\min(L[u], u-h)$ is a stronger score (rank correlation 0.38; band \$1.13 against \$1.97), although plain split remains narrower (\$0.93). Mondrian buckets give the narrowest band (\$0.85) and the best worst-bucket coverage (0.86 against 0.66). All labels are synthetic; we claim no market coverage.**

***Index Terms*—physics-informed neural networks, American options, conformal prediction, uncertainty quantification, no-arbitrage**

I. Introduction

A trader who receives a model price also needs to know how far that price may be from the truth. Without an uncertainty band, that price is not yet a usable risk number. An American contract adds a second difficulty. Its holder can exercise early, the exercise boundary has to be found as part of the solution, and no closed-form price exists [1, 2]. Trees and Monte Carlo produce accurate reference values [3, 4]. They cannot tell us, however, the error of a fast surrogate at an unseen point. Learned pricers bring two further issues. Regressors trained on labelled prices need many labels and can break calendar or convexity conditions. PINNs require no labels [5], yet their pointwise error remains unknown until it is measured.

This paper puts distribution-free conformal bands [6, 7] around PINN pricers and asks which difficulty score should scale the band. Our contributions are as follows.

1) A controlled study of split, residual-normalised and Mondrian conformal prediction on a European PINN (N1), its no-physics control (N0), and an American PINN (N2), over 10 training seeds and 20 calibration splits each.
2) Evidence that Mondrian buckets give the most even coverage, which we see as the main practical result. The raw PDE residual $|L[u]|$ is a poor difficulty score for the American put. The complementarity residual $|\min(L[u], u-h)|$ is a better one, but it does not beat plain split on width once the PINN is unbiased.
3) A diagnosis and fix of an upward bias in the American PINN: the product form of the complementarity penalty barely penalised a positive residual.
4) A labelled-baseline comparison and a static-arbitrage check of PINN prices and of interval edges, both reported without repair.

We state the limits at the outset. All labels are synthetic and exact. After the fix, our American PINN keeps an upward bias of \$0.22 on a \$100 strike, and it breaks the calendar check on 16.8% of pairs. We report no coverage on market quotes.

II. Related Work

Reference pricers. Our European ground truth is the closed-form price of [8]. The no-arbitrage conditions we test come from Merton's bounds [1]. For the American put we use the recombining tree of [3], which compares holding and exercising at every node. Two other routes exist: a quadratic analytic approximation [9], and simulation in which a regression estimates the value of continuing [4]. Under stochastic volatility [10], grid methods such as ADI splitting [11] are a common choice. Standard texts such as [2] cover put-call parity and early exercise.

Networks as PDE solvers. A PINN [5] is fitted by penalising, at random points, how far it is from satisfying the PDE, its boundary values and its initial or terminal value. DGM [12] is a close mesh-free relative and includes an American option among its tests. The deep BSDE solver [13] aims at PDEs in many dimensions. In finance, PINN variants have priced contracts under Black-Scholes type equations [14], with early exercise [15] and under Heston dynamics [16]. As far as we can tell, these papers measure price accuracy and do not report calibrated error bands.

Learners trained on prices. For tabular data, random forests [17, 18] and boosted trees [19] are standard, well-tested choices. An input-convex network [20] is convex in chosen inputs by design, and option prices are convex in the underlying.

Distribution-free bands. When the data are exchangeable, a conformal set contains the truth with a stated probability for any sample size [6]. A cheap variant fits the model once and sets the band width from residuals on points kept aside [21]; Lei et al. [7] study what this variant guarantees for regression. When each residual is scaled by an estimate of local difficulty, the band width can change from point to point [22]. CQR [23] obtains a similar effect from quantile models, and Mondrian regressors [24] repeat the calibration inside each group of inputs. Angelopoulos and Bates [25] give an accessible overview.

Gap. We found no study that uses the physics residual of a PINN pricer as the conformal difficulty score, and none that asks whether that residual is valid for a free-boundary problem.

III. Method

A. Pricing problem

We write S for the spot price and K for the strike. Time to expiry is $\tau = T - t$. The interest rate, dividend yield and volatility are r, q and σ. With moneyness $m = S/K$ and price per unit strike $u(m,\tau) = V/K$, the pricing operator of [1, 8] reads

$$L[u] = u_\tau - (r-q)\,m\,u_m - \tfrac{1}{2}\sigma^2 m^2 u_{mm} + r\,u. \tag{1}$$

A European put solves $L[u] = 0$ with $u(m,0) = h(m) = \max(1-m,0)$, $u(0,\tau) = e^{-r\tau}$ and $u(m_{\max},\tau) = 0$. The American put solves the linear complementarity problem (LCP)

$$L[u] \geq 0, \qquad u \geq h, \qquad L[u]\,(u-h) = 0. \tag{2}$$

In the exercise region $u = h = 1-m$, and (1) gives $L[h] = r - q\,m$, which equals $r > 0$ when $q = 0$. So a positive $L[u]$ there is correct, not an error.

B. Networks and losses

Each PINN is a tanh multilayer perceptron with 4 hidden layers of 64 units on input (m,τ). The residual contains u_{mm}. A ReLU network is piecewise linear, so this term would vanish except on a set of measure zero; tanh avoids the problem. We obtain u_m, u_{mm} and u_τ with the automatic differentiation of PyTorch [26]. Let $\overline{(\cdot)}$ denote a mean over points resampled every epoch across 3000 interior points drawn on $[0,3]\times[0,1]$, together with 400 terminal points (half near $m = 1$) and 200 boundary points. The European loss is

$$\mathscr{L}_{\mathrm{E}} = \lambda_p \overline{L[u]^2} + 3\,\overline{(u-h)^2}_{\tau=0} + \overline{(u-g)^2}_{\partial}, \tag{3}$$

where g is the boundary value. N1 uses $\lambda_p = 1$. N0 uses $\lambda_p = 0$, so it fits only the payoff and the edges; it is the control. The American loss N2 replaces the PDE term with three LCP penalties,

$$\overline{\mathrm{relu}(-L[u])^2} + \overline{\mathrm{relu}(h-u)^2} + 10\,\overline{\min(L[u],u-h)^2}, \tag{4}$$

plus the same terminal and boundary terms, except that the put takes its American value $u(0,\tau) = 1$ at $m = 0$. All runs use fixed weights. Training uses Adam with learning rate 2×10^{-3}, halved every 2000 of 6000 epochs. The PINNs see no labelled price.

C. Labelled baselines

Table I lists ten labelled regressors from an earlier project run. We retrain its PINN row with the N2 loss above. They learn $(m,\tau)\mapsto u$ from 500-step binomial labels [3]. The four classical models use scikit-learn [27] and the six networks use PyTorch.

D. Conformal intervals

Let $\hat{u}$ be a PINN prediction and $R(m,\tau)$ a signed residual. On n calibration points with exact labels we compute

$$s_i = \frac{|u_i - \hat{u}_i|}{1+\kappa|R_i|}, \qquad \kappa = \frac{1}{\mathrm{median}_i\,|R_i|}. \tag{5}$$

With $\kappa = 0$ this is split conformal [21]; with $\kappa > 0$ it is a normalised score [22]. Sort the scores and take q as the one in position

$$k = \lceil (n+1)(1-\alpha) \rceil, \tag{6}$$

counting from the smallest. This rank is the finite-sample correction of split conformal [7]. At a test point, the dollar band is

$$K\hat{u} - w \;\leq\; V \;\leq\; K\hat{u} + w, \qquad w = Kq\,(1+\kappa|R|). \tag{7}$$

When calibration and test points are exchangeable, this band misses the true price with probability at most α [7]. We compare four scores: split ($\kappa = 0$); normalised with $R = L[u]$; for N2 only, normalised with the LCP residual $R = \min(L[u], u-h)$, which vanishes exactly when (2) holds; and Mondrian [24]. Mondrian stores a separate q in each of six cells formed by three moneyness ranges (below 0.9, from 0.9 to 1.1, above 1.1) crossed with two tenor ranges (below 0.4 and from 0.4 up).

IV. Experiments

A. Setup

Every run uses a 100-dollar strike, no dividends, a 3% rate and 20% volatility ($K = 100$, $q = 0$, $r = 0.03$, $\sigma = 0.2$). N0, N1 and N2 are each trained with seeds 0 to 9 (30 runs). Each seed gets its own pool of 4000 points, $m \sim U[0.6, 1.4]$ and $\tau \sim U[0.05, 1]$. European labels come from the closed form;

TABLE I

AMERICAN PUT: ACCURACY OF THE LABELLED REGRESSORS (MAE IN \$, $K = 100$). LABELLED ROWS: EARLIER PROJECT RUN ON A SEPARATE 25×9 TEST GRID, MEAN OF 2 SEEDS. PINN ROW: N2 RETRAINED WITH THE LOSS OF SECTION III ON THE SAME GRID. "BOX": LABELS ONLY FOR $m \in [0.85, 1.15]$, $\tau \leq 0.5$, SCORED INSIDE AND OUTSIDE THE BOX. VIOLATION RATES ARE FINITE-DIFFERENCE CHECKS ON THE FULL-LABEL GRID AT 3000 LABELS.

	Full labels		Box, 3000		Viol. rate	
Model	n=50	3000	in	out	convex	calendar
Polynomial ridge (deg. 6)	0.441	0.425	0.026	142.81	0.29	0.21
Kernel ridge (RBF, CV)	0.432	0.005	0.003	67.06	0.09	0.06
Random forest [17, 18]	0.961	0.047	0.057	3.98	0.20	0.02
Gradient boosting [19]	1.328	0.108	0.062	3.99	0.28	0.31
MLP, tanh, 4x64	0.344	0.229	0.168	2.62	0.19	0.18
MLP, ReLU, 4x64	0.254	0.027	0.027	2.46	0.13	0.13
Residual MLP, SiLU	0.190	0.099	0.076	5.06	0.10	0.26
Fourier-feature MLP	8.318	0.053	0.034	8.40	0.34	0.34
Partial ICNN [20]	1.360	0.599	0.291	3.45	0.00	0.20
BS-embedded IV net	0.050	0.009	0.022	0.13	0.05	0.05
PINN N2 (no labels)	0.231	0.231	0.209	0.23	0.16	0.20

American labels come from a CRR tree with 500 steps. For each of 20 random splits, the first $n_{\text{cal}} = 800$ points calibrate and the next 2000 test; $n_{\text{cal}} = 400$ gave the same picture. We use $\alpha \in \{0.10, 0.05\}$. We average over splits within a seed, then over seeds. "Worst bucket" is the lowest of the six bucket coverages in a split, averaged the same way. The protocol was written before any coverage was computed and was not changed. We later fixed the N2 loss on a separate validation grid and wrote a second protocol before computing any coverage of the fixed model. All runs used CPU only, on an 8-vCPU Intel Xeon cloud machine with one thread per run and 8 runs in parallel. One run took 157 s (N0), 167 s (N1) and 165 s (N2) on average; the 10 N2 runs took 5 min 20 s in total. All conformal evaluations took 33 s. Numerical code used NumPy [28].

B. Labelled baselines

Table I confirms that labels win inside the labelled region. At 3000 labels, kernel ridge reaches \$0.005 against \$0.231 for the PINN, and eight of ten labelled models outperform it, whereas at 50 labels only two of ten do. Once labels are restricted to a strip near the money, kernel ridge and polynomial ridge climb to \$67.06 and \$142.81 outside that strip, while the PINN remains at \$0.23, close to its full-grid error. The PINN fails the finite-difference calendar check in 20% of cases and sits mid-pack among the labelled models. Outside the box the best model is the Black-Scholes-embedded network (\$0.13), because the formula itself is part of its architecture; the PINN is next.

C. Point accuracy and the residual signal

Table II confirms that the PDE term drives accuracy, since N1 reaches an MAE of \$0.25 whereas the N0 control sits at \$4.42. N2 has an MAE of \$0.29 and a bias of +\$0.22: it prices 83.7% of points above the binomial reference. The residuals behave differently by model. For N0 the raw residual $|L[u]|$ tracks the error ($\rho = 0.56$), because N0 never fits the PDE. For N1 it is weak ($\rho = 0.15$). For N2 it carries no signal ($\rho = 0.01$), while the LCP residual has $\rho = 0.38$. Fig. 1 shows the pattern for one seed. The raw residual (median 0.0071) shows no trend with the price error. One reason is structural: in the exercise region the exact solution has $L[u] = r$, so a large $|L[u]|$ there is not an error. The LCP residual removes this part. It rises with the error mainly in the exercise region, where it equals the gap $u - h$.

TABLE II

PINN POINT ACCURACY ON THE CALIBRATION/TEST POOLS (MEAN OF 10 SEEDS, \$). BIAS IS THE MEAN SIGNED ERROR; "ABOVE" IS THE SHARE OF POINTS PRICED ABOVE THE REFERENCE; ρ IS THE SPEARMAN RANK CORRELATION OF THE ABSOLUTE RESIDUAL WITH THE ABSOLUTE ERROR.

Model	MAE	Max	Bias	Above	$\rho_{\lvert L\rvert}$	ρ_{LCP}	Train s
N0 (Eur., no PDE)	4.42	14.50	+4.12	0.876	0.56	n/a	157
N1 (Eur., PDE)	0.25	0.67	+0.04	0.629	0.15	n/a	167
N2 (Amer., LCP)	0.29	0.84	+0.22	0.837	0.01	0.38	165

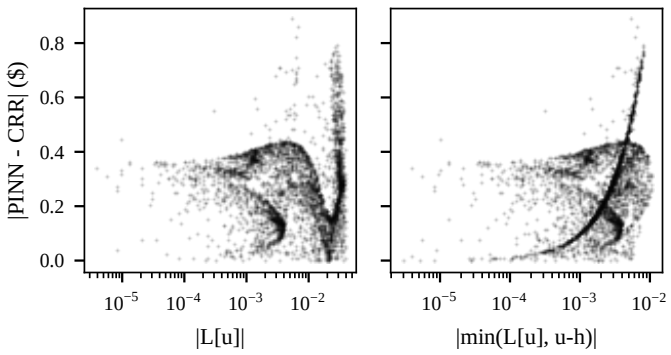


Fig. 1. N2, seed 0, 4000 pool points. Absolute price error against the raw residual $|L[u]|$ (left) and the LCP residual (right). The raw residual shows no trend with the error; the LCP residual rises with it mainly in the exercise region (the curved branch).

D. Conformal bands on PINN pricers

Table III gives the main result. Marginal coverage is at its nominal level for every model and score, within 0.005. The same held for N2 before we corrected its bias, so the finite-sample guarantee did its job and the bias appeared as width (split band \$10.05). Adding the PDE term shrinks the split band from \$19.43 (N0) to \$0.94 (N1) at $\alpha = 0.10$, roughly 20 times. The choice of difficulty score matters. Normalising by $|L[u]|$ widens both the N1 band (\$1.18 against \$0.94) and the N2 band (\$1.97 against \$0.93), and it helps only N0, whose residual is large where that model fails. For N2 the LCP residual narrows the band relative to $|L[u]|$ (\$1.13 against \$1.97) but not relative to split (\$0.93), and its worst bucket (0.64) sits below split's (0.66). Mondrian yields the narrowest bands together with the best worst-bucket coverage (0.86 against 0.66 for split on N2). Fig. 2 shows that split under-covers deep in-the-money short tenors (0.69) and over-covers near-the-money long tenors (0.999), whereas Mondrian holds each bucket between 0.901 and 0.906 on average. Fig. 3 shows the bands on one slice: they contain the reference and

TABLE III
CONFORMAL BANDS, $n_{\text{cal}} = 800$, $n_{\text{test}} = 2000$, MEAN OVER 10 SEEDS × 20 SPLITS. WIDTH IN \$. SEED-TO-SEED STD OF COVERAGE IS AT MOST 0.004.

		$\alpha = 0.10$			$\alpha = 0.05$		
Model	Score	Cov.	Width	Worst	Cov.	Width	Worst
N0	Split	0.900	19.43	0.61	0.949	22.78	0.79
N0	Norm. $\lvert L[u]\rvert$	0.900	16.18	0.66	0.951	18.61	0.82
N0	Mondrian	0.904	13.59	0.86	0.954	14.81	0.92
N1	Split	0.900	0.94	0.63	0.950	1.07	0.78
N1	Norm. $\lvert L[u]\rvert$	0.898	1.18	0.74	0.949	1.42	0.85
N1	Mondrian	0.903	0.80	0.86	0.954	0.87	0.92
N2	Split	0.901	0.93	0.66	0.951	1.04	0.78
N2	Norm. $\lvert L[u]\rvert$	0.901	1.97	0.65	0.950	2.24	0.81
N2	Norm. LCP	0.900	1.13	0.64	0.951	1.30	0.80
N2	Mondrian	0.904	0.85	0.86	0.954	0.92	0.92

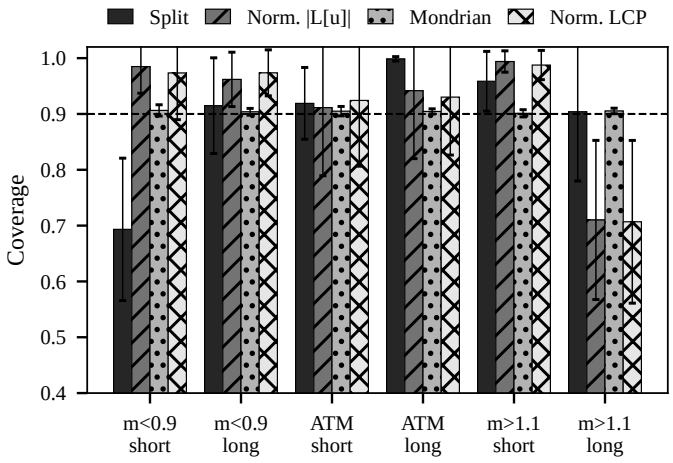


Fig. 2. N2, $\alpha = 0.10$, $n_{\text{cal}} = 800$: coverage per moneyness and tenor bucket (short: $\tau < 0.4$), mean and std over 10 seeds. Dashed line: target 0.90. Only Mondrian is close to target in every bucket.

stay close to it, but the lower edges fall below zero far out of the money.

E. Synthetic heteroscedastic check

To isolate the score, we reuse a synthetic set with a known mean, noise standard deviation $0.01 + 0.12|m-1|$, difficulty $|m-1|$, $n_{\text{cal}} = 800$, $n_{\text{test}} = 2500$ and 50 seeds (Table IV). Normalisation narrows the band (0.127 against 0.145 at $\alpha = 0.10$) and raises coverage far from the money from 0.813 to 0.854. The fitted κ is 4.0, while the true ratio of slope to intercept is 12, so the adaptation is only partial. With homoscedastic noise, split coverage is 0.903 ($\alpha = 0.10$) and 0.953 ($\alpha = 0.05$).

F. Static arbitrage of prices and band edges

We run the project's arbitrage checker on a put chain with $S = 100$, strikes 60 to 140 in steps of 2.5 and ten expiries from 0.1 to 1 year, with tolerance 10^{-6} (Table V). Bands come from split 0, $n_{\text{cal}} = 800$, $\alpha = 0.10$. The PINN prices break strike convexity in 12% of checkable triples for both N1 and N2. The N2 point prices also break the calendar check in 16.8% of pairs, and 2.4% of these prices are negative. A constant

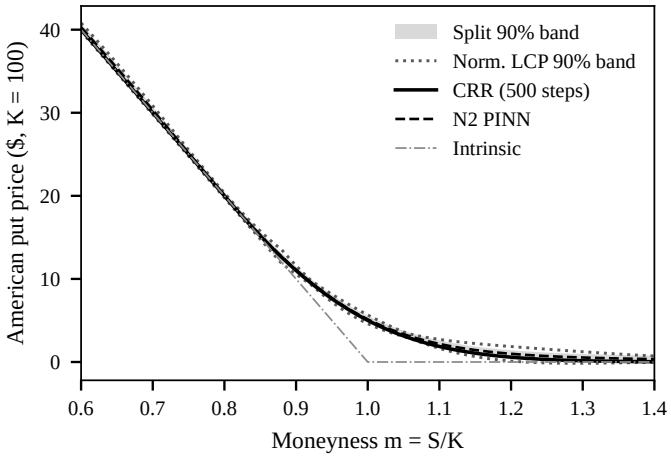


Fig. 3. American put at $\tau = 0.5$, seed 0, $\alpha = 0.10$. The split band (grey) and the LCP-normalised band (dotted) both contain the binomial price. The PINN sits a little above the reference, most clearly out of the money, and the lower edges drop below zero far out of the money.

TABLE IV
SYNTHETIC HETEROSCEDASTIC DATA, 50 SEEDS. WIDTHS ARE IN u UNITS. NEAR: $|m-1| \le 0.25$; FAR: $|m-1| > 0.25$.

α	Score	Cov.	Width	Near	Far
0.10	Split	0.899	0.145	0.985	0.813
0.10	Normalised	0.899	0.127	0.945	0.854
0.10	Mondrian	0.903	0.139	0.955	0.851
0.05	Split	0.949	0.184	0.996	0.902
0.05	Normalised	0.950	0.155	0.978	0.923
0.05	Mondrian	0.953	0.175	0.981	0.925

split band keeps the convexity and calendar rates of the point price, since it only adds a term linear in K, but its lower edge raises monotonicity violations from 0.039 to 0.053. Variable-width bands introduce further violations, and roughly a fifth of lower edges turn negative. The calendar check is not a valid condition for European puts, since the exact Black-Scholes chain fails it in 27% of cases, and we apply no repair.

V. DISCUSSION

Keeping the PINN despite its MAE. The PINN needs no labels and degrades gently outside any labelled region, where kernel and polynomial ridge fail (Table I), and it also exposes a residual for free. Once a conformal layer is added, even a biased PINN still yields bands with valid marginal coverage. Bias is paid for in width, and improving the PINN therefore narrows the bands at once, which both N1 against N0 and N2 before against after its fix demonstrate (\$10.05 against \$0.93).

Choice of residual. The residual should measure distance from the conditions that define the solution. For a European option that is $L[u] = 0$. For an American option it is the LCP (2), and $|L[u]|$ is not zero at the true solution in the exercise region. The LCP residual corrects this mismatch and serves as a stronger error proxy than $|L[u]|$. After the bias is fixed, however, no residual scaling beats plain split on width, and

TABLE V
STATIC-ARBITRAGE VIOLATION RATES ON A SYNTHETIC PUT CHAIN (MEAN OF 10 SEEDS). PUT-CALL PARITY IS NOT APPLICABLE TO A PUT-ONLY CHAIN.

Series	Monot.	Convex	Calendar	Price<0
CRR reference (Amer.)	0.000	0.000	0.000	0.00
BS reference (Eur.)	0.000	0.000	0.269	0.00
N1 point price	0.036	0.121	0.368	0.09
N2 point price	0.039	0.121	0.168	0.02
N2 split, lower edge	0.053	0.121	0.168	0.20
N2 split, upper edge	0.033	0.121	0.168	0.00
N2 norm. LCP, lower	0.113	0.266	0.229	0.20
N2 norm. LCP, upper	0.042	0.226	0.240	0.00
N2 Mondrian, lower	0.051	0.145	0.164	0.18
N2 Mondrian, upper	0.033	0.148	0.180	0.00

Mondrian buckets continue to deliver the most even coverage, so we leave a joint treatment of the two ideas to future work.

The N2 bias. Our first N2 overpriced by \$3.13 on average, and we traced the bias to the complementarity penalty. Its product form $L[u](u-h)$ is tiny where $u-h$ is small, so a network with $L[u]>0$ in the continuation region paid almost nothing. We replaced it with $\min(L[u],u-h)$ at weight 10 and set the American value 1 at $m=0$. Changing the boundary alone barely moved the bias. We selected the fix on a separate validation grid rather than on calibration or test points. The fixed N2 still carries a bias of +\$0.22, largest for deep out-of-the-money long tenors (+\$0.35), breaks the calendar check more often than before, and sits below intrinsic value by more than one cent on 11% of pool points.

Limits of the market snapshot. We keep one Yahoo Finance snapshot of 4054 quotes dated 13 Aug 2026 for SPX, NDX, NVDA and AAPL, and because that single snapshot cannot support a time-ordered calibration and test split we report no market coverage, while real quotes further carry bid-ask noise and fail exchangeability across dates.

VI. CONCLUSION

We fitted split, normalised and Mondrian conformal bands on European and American PINN pricers, and marginal coverage held for every model both before and after we corrected the American PINN bias. The PDE loss cut the European band by about 20 times, and with 3000 labels most labelled regressors still remain more accurate in-sample.

A useful difficulty score must fit the problem, so the raw PDE residual is a weak score for the American put while the complementarity residual is stronger and Mondrian buckets remain the most useful choice in practice. Future work will clear the calendar violations and obstacle slack of N2, assemble a dated market panel for real coverage tests, project band edges onto arbitrage-free prices, and place a band on the exercise boundary $S^*(\tau)$.

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